

Results

01 • ONE-WAY ANOVA + POST-HOC

do 3+ groups differ, and which pairs?

Table 1. Descriptives by group

Group	N	M	SD
control	20	32.48	7.47
drug_a	20	33.84	6.91
drug_b	20	37.81	7.79

Table 2. ANOVA

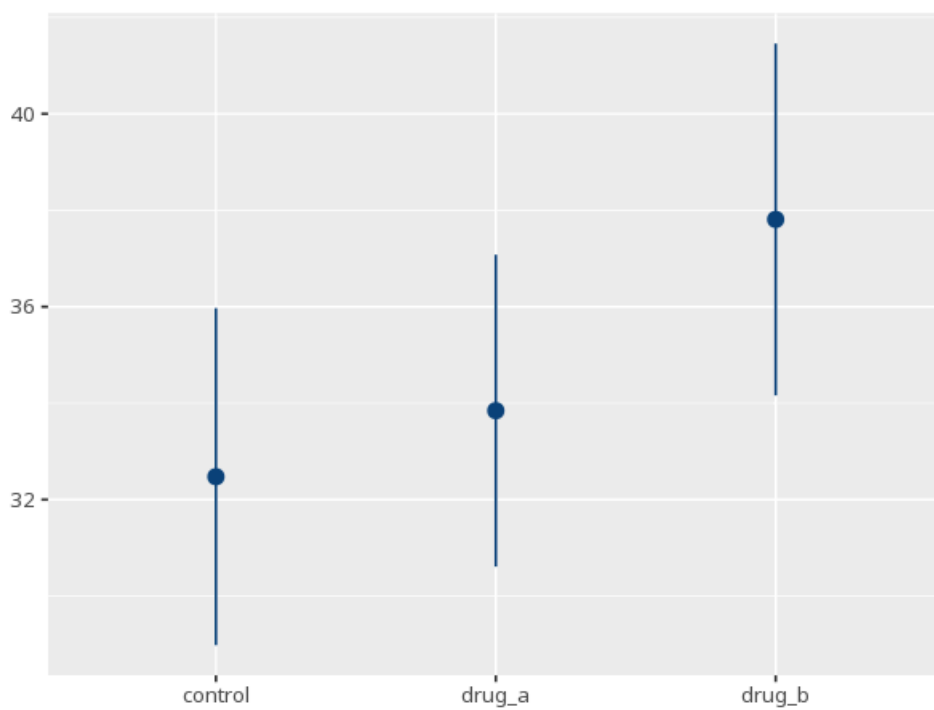
Source	SS	df	MS	F	p	η^2 [95% CI]
Between	307.07	2	153.53	2.81	.069	0.09 [0.00, 1.00]
Within	3119.95	57	54.74			

Table 3. Post-hoc comparisons

Pair	M _{diff}	SE	P _{adj}	95% CI
control - drug_a	-1.37	2.34	.828	[-7.00, 4.26]
control - drug_b	-5.33	2.34	.067	[-10.96, 0.29]
drug_a - drug_b	-3.96	2.34	.216	[-9.59, 1.66]

assumption checks: Levene's (equal variances) & normality of residuals. (Levene F=0.02, p=.978 • Shapiro W=0.96, p=.059)

Figure. Group means



How to read this test

The F and its p tell you whether the groups differ overall; η^2 (or ω^2) is the effect size. If significant, the post-hoc table shows which specific pairs differ, with multiplicity-adjusted p-values. Your significance threshold (α) is 0.05.

APA template: A one-way ANOVA gave $F(2,57)=2.81$, $p = .069$, $\eta^2=.09$ [.00, 1.00]. Tukey HSD post-hoc tests showed...

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